

Investigate the stability of the torque (issue #18), revised

September 2026

1 Short summary

This report checks the torque sampling used for `motor1` and `motor2`. The sampling depends on both `PointPer` and N_{per} , which together determine the number of torque samples. The aim is to check the stability of `TorAvg` and `TorRippleRms` and to find a practical sampling resolution considering both accuracy and computation time.

2 Method

The transient settings are

$$\text{StopTime} = \frac{N_{\text{per}}}{f}, \quad \text{TimeStep} = \frac{1}{f(\text{PointPer} - 1)}$$

The number of returned torque samples is therefore

$$n_{\text{points}} \approx N_{\text{per}}(\text{PointPer} - 1) + 1$$

Here, $N_{\text{per}} = 1/6$ for `motor1` and $N_{\text{per}} = 1/10$ for `motor2`. Thus, the same `PointPer` gives different numbers of torque samples for the two motors.

Three rotor parameterizations were tested for each motor: `HacklGenerator_OneLambda`, `HacklGenerator_SixLambdas`, and `HacklGenerator_3BrokenLines`. One fixed design was used for each parameterization during the sweep. `PointPer` was varied from 51 to 1001 in steps of 50. All other settings were kept fixed. `TorAvg` and `TorRippleRms` were evaluated with `analyze_results()`.

For each motor and rotor parameterization, the mean of the five largest `PointPer` cases, from 801 to 1001, was used as the reference value. The relative error is

$$\varepsilon_q = \frac{|q - q_{\text{ref}}|}{|q_{\text{ref}}|} \times 100\%$$

3 PointPer sensitivity

Figures 1 and 2 show the relative errors for the three rotor parameterizations over the full tested range from `PointPer` = 51 to 1001. Convergence is taken as the first value after which both errors remain below the specified limit.

At the current `PointPer` = 101, the largest errors in `TorAvg` and `TorRippleRms` are 1.68% and 6.07% for `motor1`, and 1.26% and 8.23% for `motor2`. The torque ripple is more sensitive to the sampling resolution.

For `motor1`, the 1% limit is reached from `PointPer` = 301 for `OneLambda` and `3BrokenLines`, and from `PointPer` = 201 for `SixLambdas`. For `motor2`, the corresponding values are 451, 451, and 401. For the 0.5% limit, the conservative values are `PointPer` = 501 for `motor1` and 601 for `motor2`.

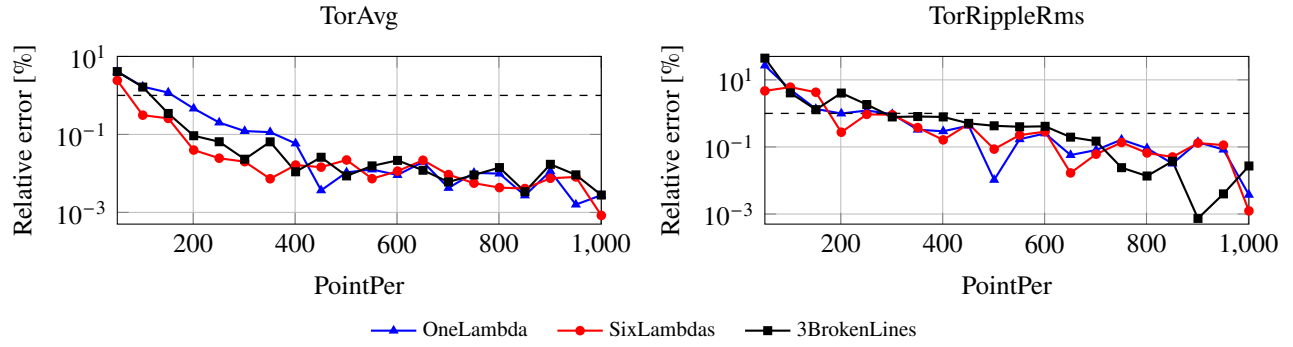


Figure 1: Relative errors for the three rotor parameterizations of `motor1` with increasing `PointPer`. The dashed line indicates 1%.

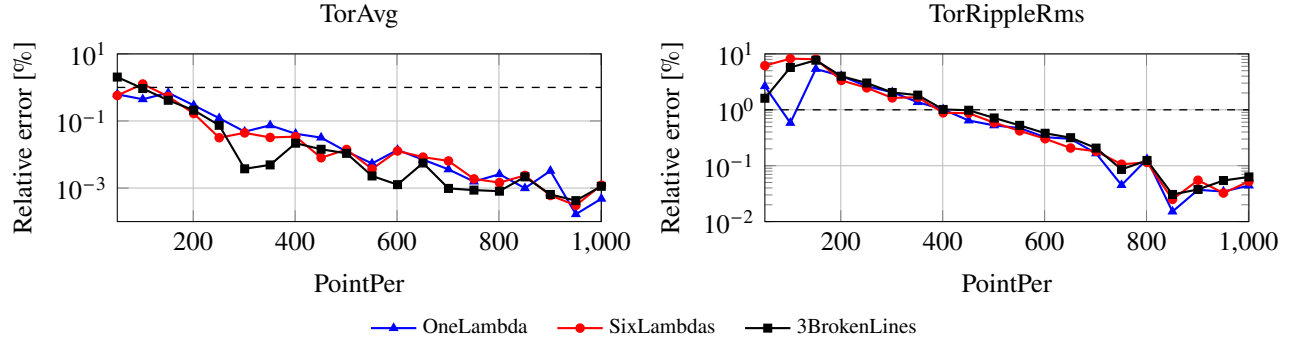


Figure 2: Relative errors for the three rotor parameterizations of `motor2` with increasing `PointPer`. The dashed line indicates 1%.

Figure 3 shows the elapsed computation time. The time increases approximately linearly with `PointPer` for all three rotor parameterizations. At the conservative 1% settings, the computation time is about 2.0 to 2.1 times the current value for `motor1`, and about 2.1 to 2.5 times the current value for `motor2`.

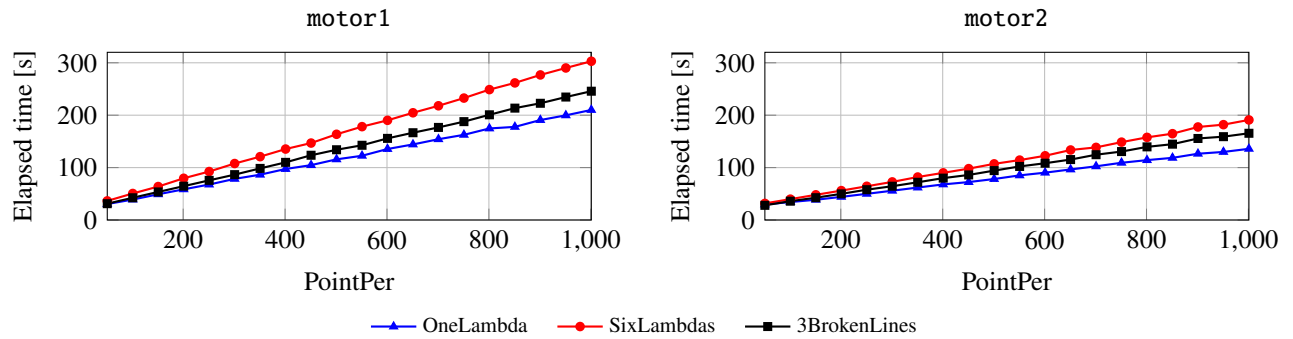


Figure 3: Elapsed computation time for the three rotor parameterizations with increasing `PointPer`.

4 Number of torque samples

The same `PointPer` gives different n_{points} because the two motors use different N_{per} . Therefore, n_{points} is used to compare the sampling convergence of the two motors. Figures 4 and 5 show the same relative errors using n_{points} as the horizontal axis.

For the 1% limit, `OneLambda`, `SixLambdas`, and `3BrokenLines` require 51, 35, and 51 samples for `motor1`. The corresponding values for `motor2` are 46, 41, and 46 samples. Thus, about 50 torque samples are sufficient

for the 1% limit for the cases tested here.

For the 0.5% limit, the required values are 60, 60, and 85 samples for `motor1`, and 56, 56, and 61 samples for `motor2`. The tighter limit shows a larger difference between the tested cases.

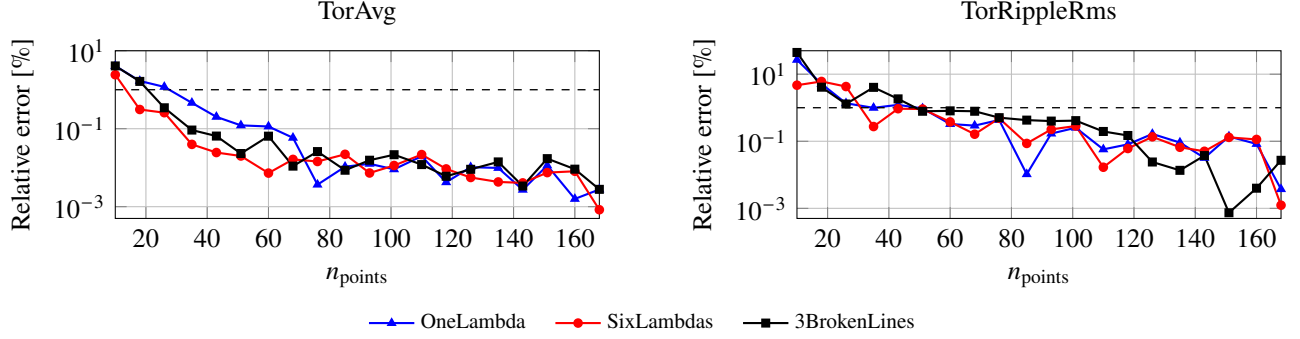


Figure 4: Relative errors for the three rotor parameterizations of `motor1` with increasing n_{points} . The dashed line indicates 1%.

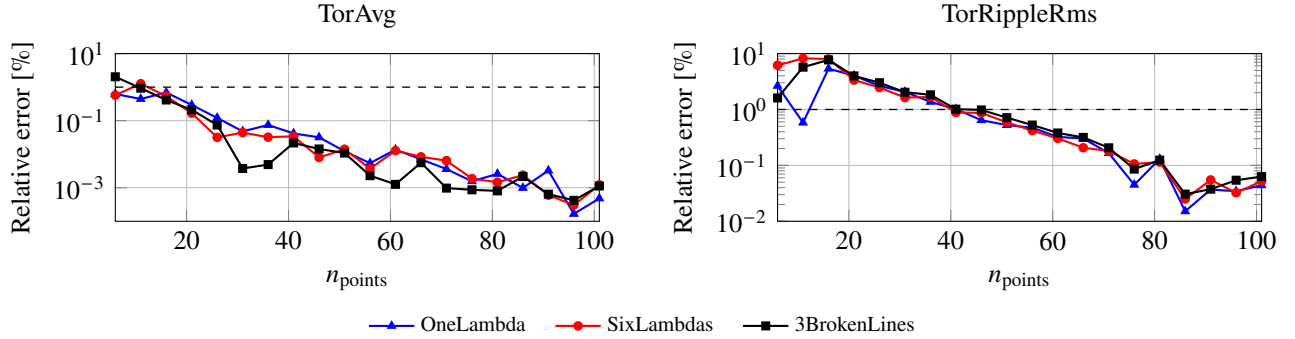


Figure 5: Relative errors for the three rotor parameterizations of `motor2` with increasing n_{points} . The dashed line indicates 1%.

5 Recommendation

For the cases tested here, n_{points} gives a more direct comparison of sampling stability between the two motors. About 50 returned torque samples are sufficient to keep both `TorAvg` and `TorRippleRms` within 1% of the reference. With the current N_{per} values, this corresponds to `PointPer` = 301 for `motor1` and `PointPer` = 451 for `motor2`. The corresponding computation time is about 2 to 2.5 times the current setting.

For a tighter 0.5% limit, the conservative settings are 85 samples for `motor1` and 61 samples for `motor2`, corresponding to `PointPer` = 501 and 601, respectively.

The current `PointPer` = 101 gives only 18 and 11 samples and is not sufficient for accurate torque ripple.